

1. Bob and Palindromes problem

```
MOD = 10**9 + 7
def modpow(a, e, mod=MOD):
    res = 1
    a %= mod
    while e > 0:
        if e & 1:
            res = (res * a) % mod
        a = (a * a) % mod
        e >>= 1
    return res
def expected_points_mod(n: int)-> int:
    inv26 = modpow(26, MOD-2)
    M = n // 2
    rem = n - 2 * M
    rpow = modpow(inv26, M)
    denom = (1 - inv26) % MOD
    inv_denom = modpow(denom, MOD-2)
    term_pairs = (2 * (1 - rpow) % MOD) * inv_denom % MOD
    term_rem = (rem * rpow) % MOD
    ans = (term_pairs + term_rem) % MOD
    return ans
if __name__ == "__main__":
    for n in [1,2,3,4,5,10]:
        print(n, expected_points_mod(n))
```

```
1 1
2 2
3 576923083
4 153846157
5 214497045
10 504844899
```

2. Soup Servings– LeetCode 808

```
from functools import lru_cache
def soupServings(N):
    if N > 4800:
        return 1.0
    n = (N + 24) // 25
    @lru_cache(None)
    def dp(a, b):
        if a <= 0 and b <= 0: return 0.5
        if a <= 0: return 1.0
        if b <= 0: return 0.0
        return 0.25 * (dp(a-4, b) + dp(a-3, b-1) + dp(a-2, b-2) + dp(a-1, b-3))
    return dp(n, n)
print(soupServings(50))
```

```
0.625
```

3. Prime numbers (Basic method)

```
def is_prime(n):
    if n <= 1:
```

```

    return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True

num = int(input("Enter a number: "))

if is_prime(num):
    print("Prime number")
else:
    print("Not a prime number")

```

Enter a number: 7
Prime number

**** $\sqrt{n}$ (root-n) optimal single-check****

```

import math
def is_prime_sqrt(n: int) -> bool:
    if n <= 1:
        return False
    if n <= 3:
        return True
    if n % 2 == 0:
        return False
    r = int(math.isqrt(n))
    d = 3
    while d <= r:
        if n % d == 0:
            return False
        d += 2
    return True

```

4. Prime Numbers(Squared root method)

```

import math

def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(math.sqrt(n)) + 1):
        if n % i == 0:
            return False
    return True

num = int(input("Enter a number: "))

if is_prime(num):
    print("Prime number")
else:
    print("Not a prime number")

```

Enter a number: 29
Prime number

